

Robotics & AI Club · Introduction

# Welcome to Robotics!

What robots are, how they work, and your first build

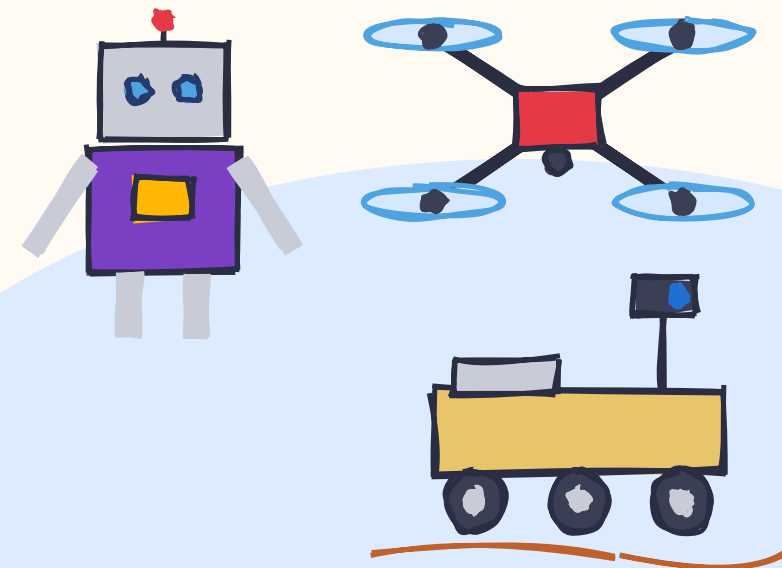
Sense, think, act

Parts of a robot

Arduino + ultrasonic sensor

Did you know?

There are more than 4 million robots working in factories around the world today.



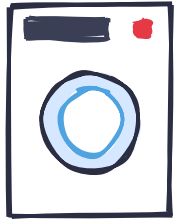


## Quick quiz

# Is it a robot? Thumbs up or down!

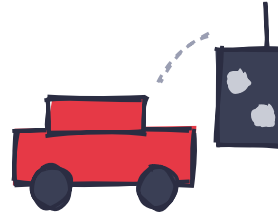


1



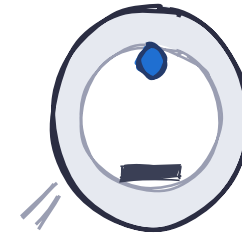
Washing machine

2



Remote-control car

3



Robot vacuum

4



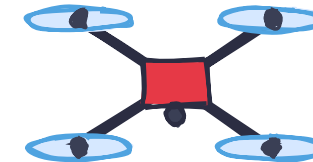
Calculator

5



Self-driving car

6



Delivery drone



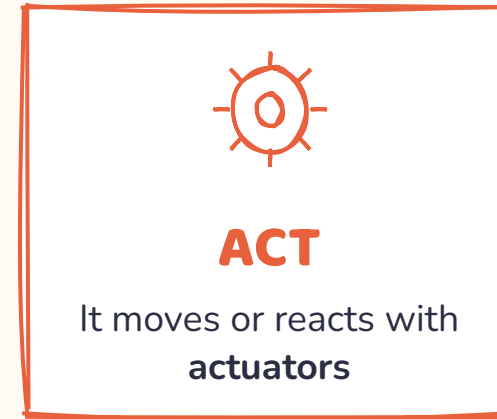
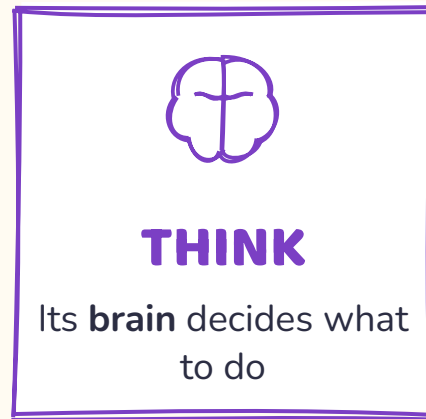
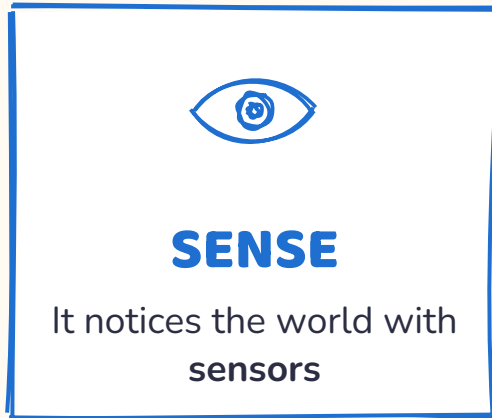
Talk with your partner: **what makes something a robot** and not just a machine?





The big idea

# A robot can SENSE, THINK and ACT



## Our definition

A robot is a **machine** that can **sense** its surroundings, **decide** what to do using a **program**, and then **act**, all **by itself**.

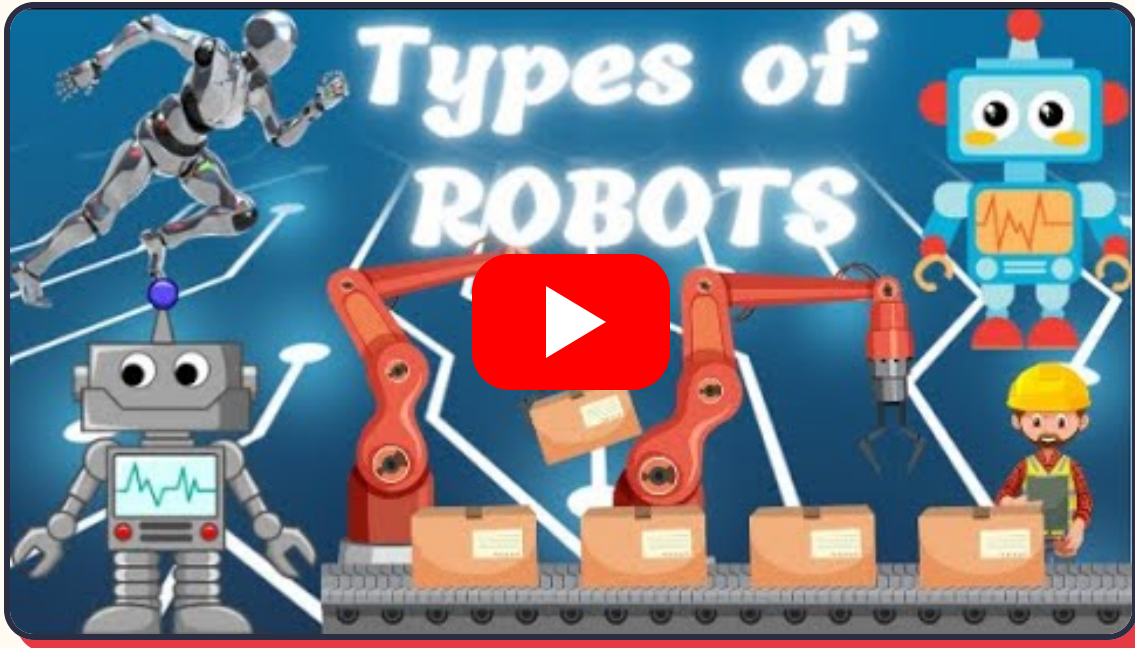
You do it too!

You **SEE** a ball coming, your brain **THINKS** 'catch!', your hands **ACT**.



Watch and think

# Video time!



## Robots for Kids: Types of Robots

Intellect Kids Classroom • about 3 minutes • [youtu.be/mm8YD99OHuA](https://youtu.be/mm8YD99OHuA)

### Watch for...

- 1 What makes a machine a **robot** and not just a machine?
- 2 Name **two jobs** robots do for people.
- 3 Which robot in the video would you most like to build?



**After the video:** Tell your partner one robot you have seen in real life. Does it **sense**, **think** and **act**?



## Answers



# So... which ones are robots?

**Sort of****Washing machine**

It follows a program and senses water level, but it cannot move around or change plans much.

**No****Remote-control car**

A person does all the thinking. No sensing, no deciding by itself.

**Yes!****Robot vacuum**

Senses walls and dirt, decides where to go, moves by itself.

**No****Calculator**

It thinks (calculates) but cannot sense the world or move.

**Yes!****Self-driving car**

Cameras and radar sense, a computer decides, motors steer and brake.

**Yes!****Delivery drone**

Uses GPS and sensors to fly a route and land on its own.

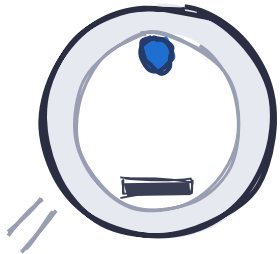


The more a machine can **sense, decide and act on its own**, the more "robot" it is. Engineers call this being **autonomous**.



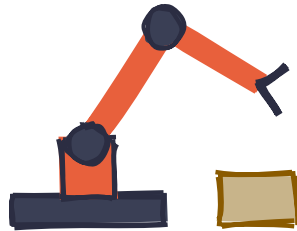
Look around you

# Robots are everywhere!



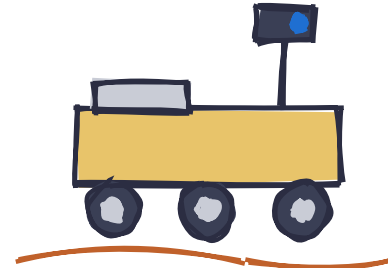
## Home helper

Cleans floors by itself



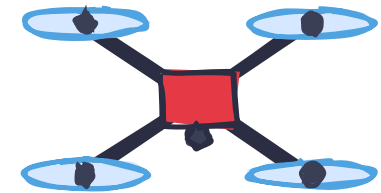
## Factory arm

Builds cars, packs boxes



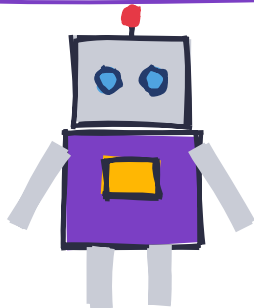
## Mars rover

Explores another planet



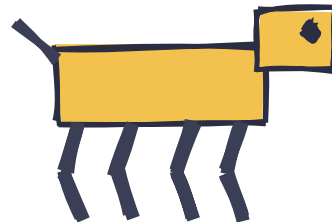
## Drone

Films, maps, delivers



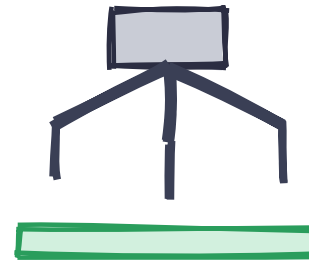
## Humanoid

Walks and talks like us



## Robot dog

Inspects dangerous places



## Surgical robot

Helps doctors operate



## Self-driving car

Drives itself on roads



## A short history

# From an idea to everywhere



**1921**

The word **robot** first appears in a Czech play by Karel Čapek. It comes from *robota*, meaning hard work.

**1997**

**Sojourner**, a rover the size of a microwave, drives on Mars.

**2021**

**Ingenuity** makes the first powered flight on another planet, Mars.

**1961**

**Unimate**, the first factory robot arm, starts work in a car factory in the USA.

**2002**

**Roomba** robot vacuums go on sale and robots enter our homes.

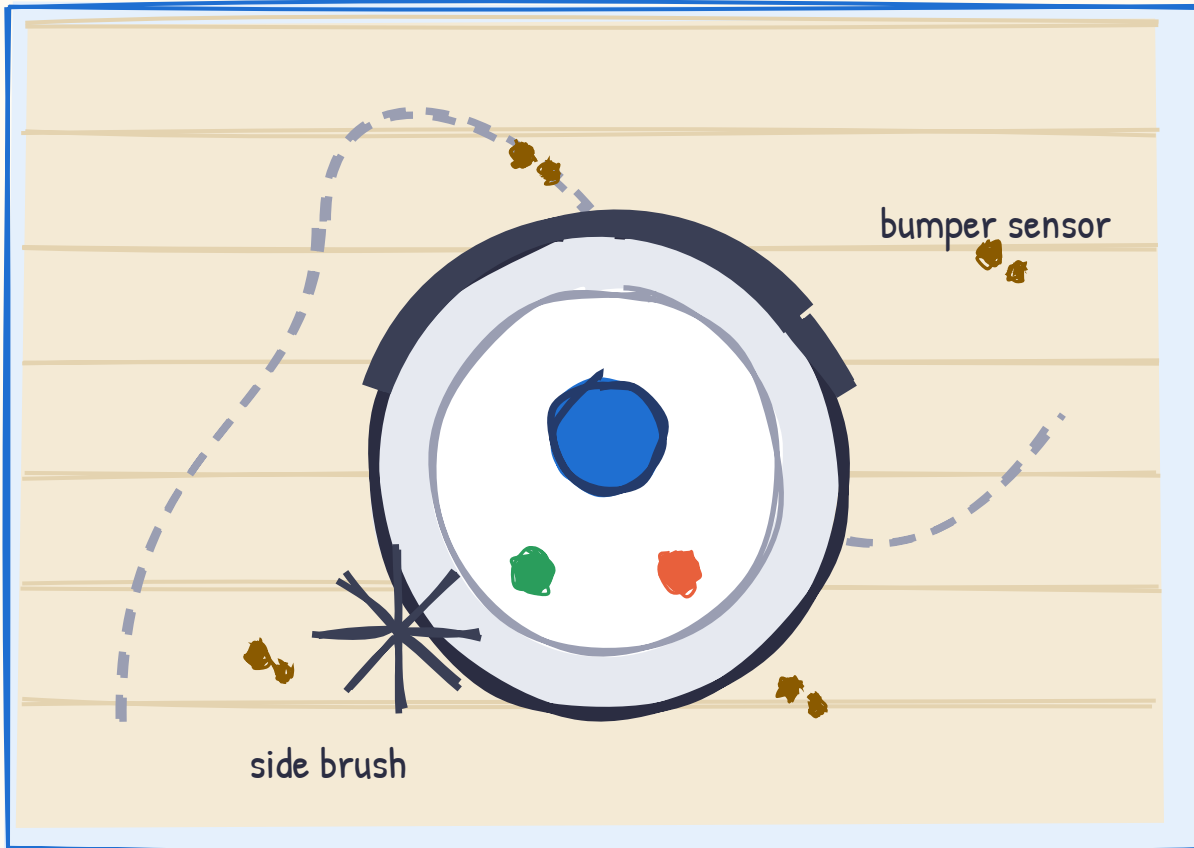
**Today**

Robots work in hospitals, farms, warehouses and space, often with **AI** inside.



Robot of the home · iRobot, USA · 2002

# Roomba



## What it does

A round robot that **vacuums your floor by itself**. It drives around furniture, cleans under beds and goes back to its **charging dock** when its battery is low.

**SENSE:** bump sensors, cliff sensors (stop falls down stairs), dirt sensors

**THINK:** plans a cleaning path

**ACT:** wheels, spinning brushes and a vacuum

## Fascinating fact

iRobot has sold tens of millions of robot vacuums. Many homes have a robot before they have any other robot!



Robot dog · Boston Dynamics, USA

# Spot



## What it does

A four-legged robot that **walks, climbs stairs and opens doors**. It inspects places that are dangerous for people, like power stations, oil rigs and building sites.

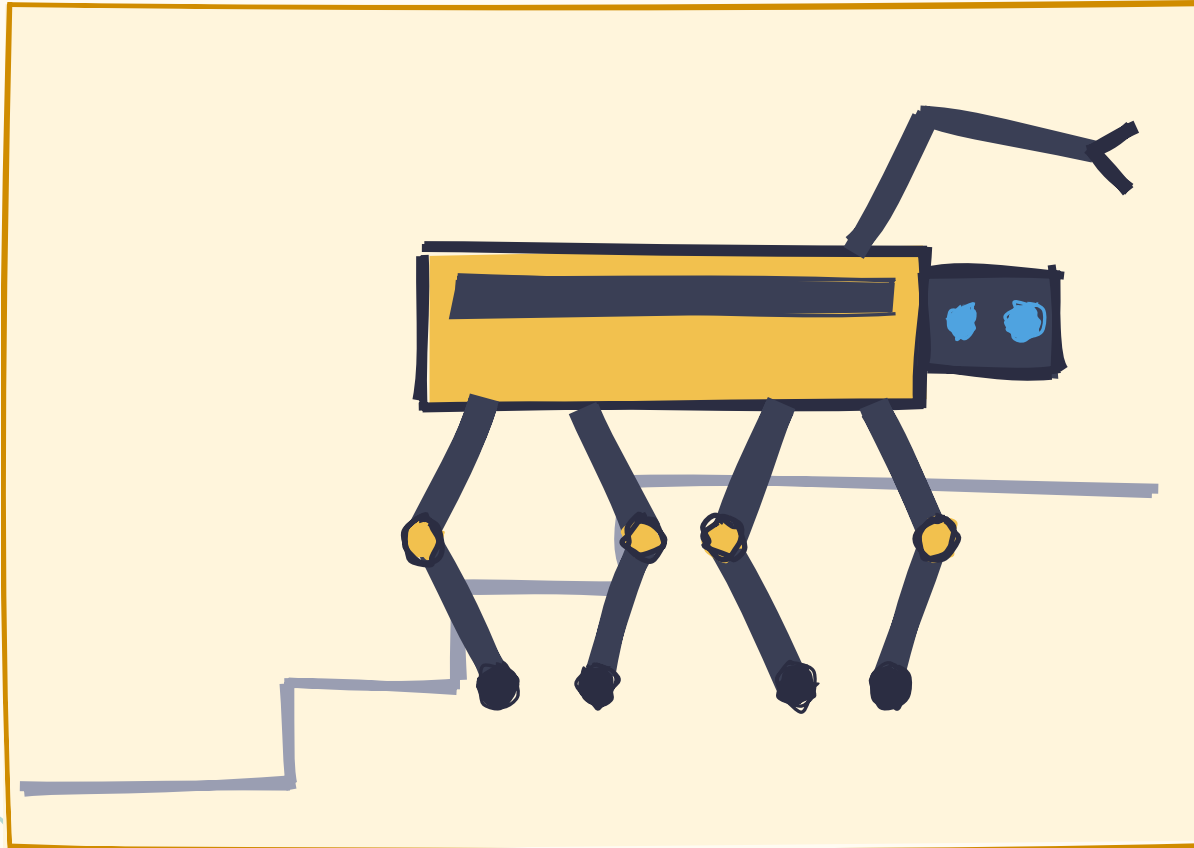
**SENSE:** cameras on every side see in 3D all around it

**THINK:** keeps its balance hundreds of times every second

**ACT:** 12 motors move its 4 legs; an optional arm can grab things

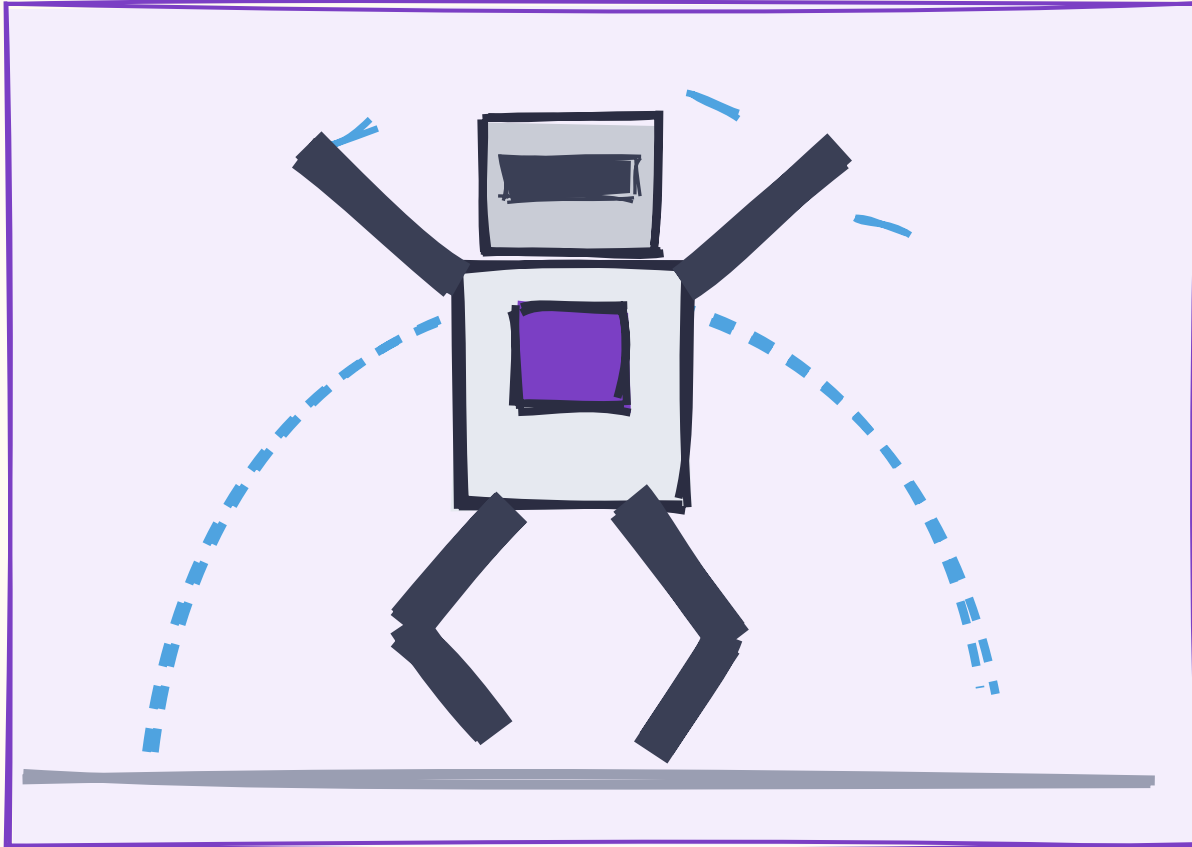
## Fascinating fact

Spot can carry about 14 kg on its back and walk on rough ground where wheels would get stuck.



Humanoid robot · Boston Dynamics, USA

# Atlas



## What it does

A robot shaped like a **human** that can run, jump and even do **backflips**. Engineers use it to learn how robots can work in places built for people.

**SENSE:** cameras and depth sensors map the ground in front of it

**THINK:** a computer plans every jump and landing

**ACT:** powerful motors in its joints move like our muscles

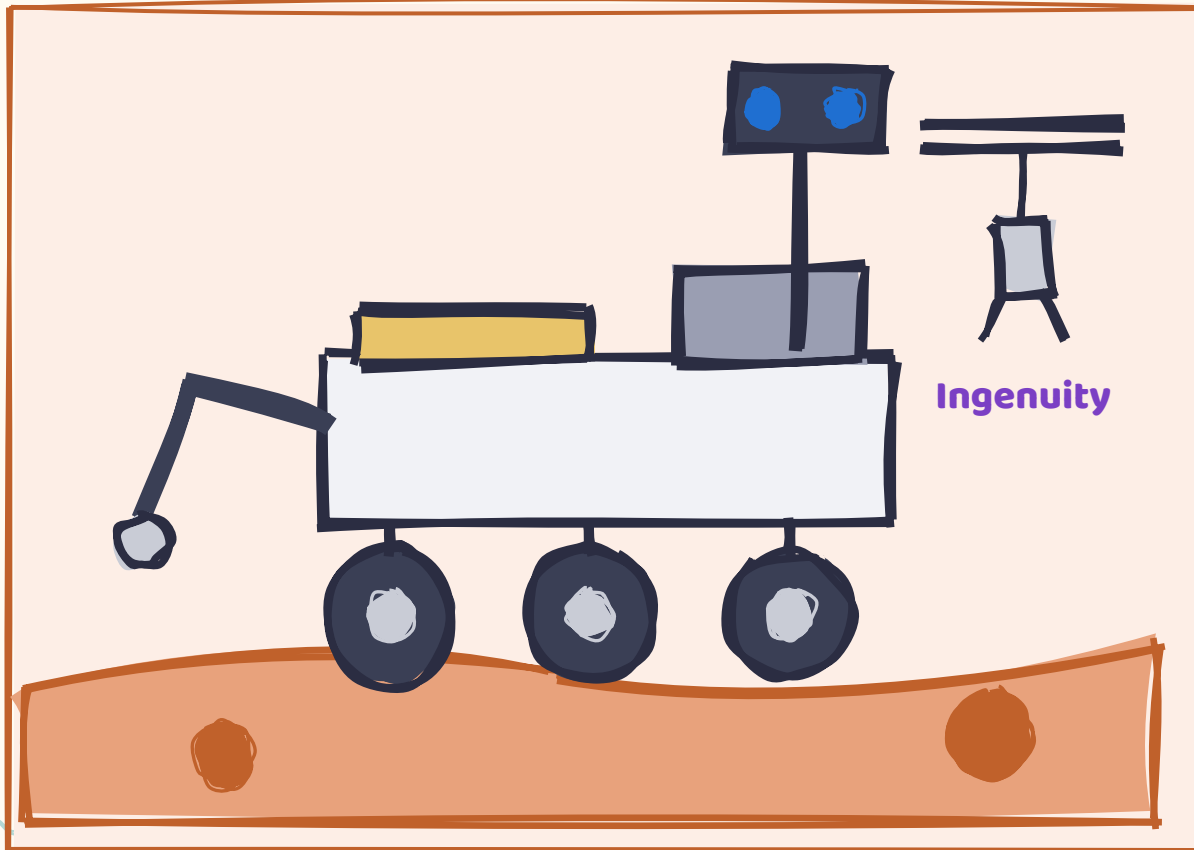
## Fascinating fact

Atlas videos doing parkour have been watched by millions of people. Balancing on two legs is one of the hardest problems in robotics!



Mars rover · NASA, USA · landed 2021

# Perseverance



## What it does

A car-sized robot **exploring Mars**. It drills rocks, collects samples and searches for signs of ancient life. It also carried **Ingenuity**, a small helicopter that flew 72 times on Mars.

**SENSE:** about two dozen cameras, a laser and science instruments

**THINK:** drives itself between targets, because radio messages take minutes to reach Earth

**ACT:** 6 wheels, a robotic arm with a drill, and sample tubes

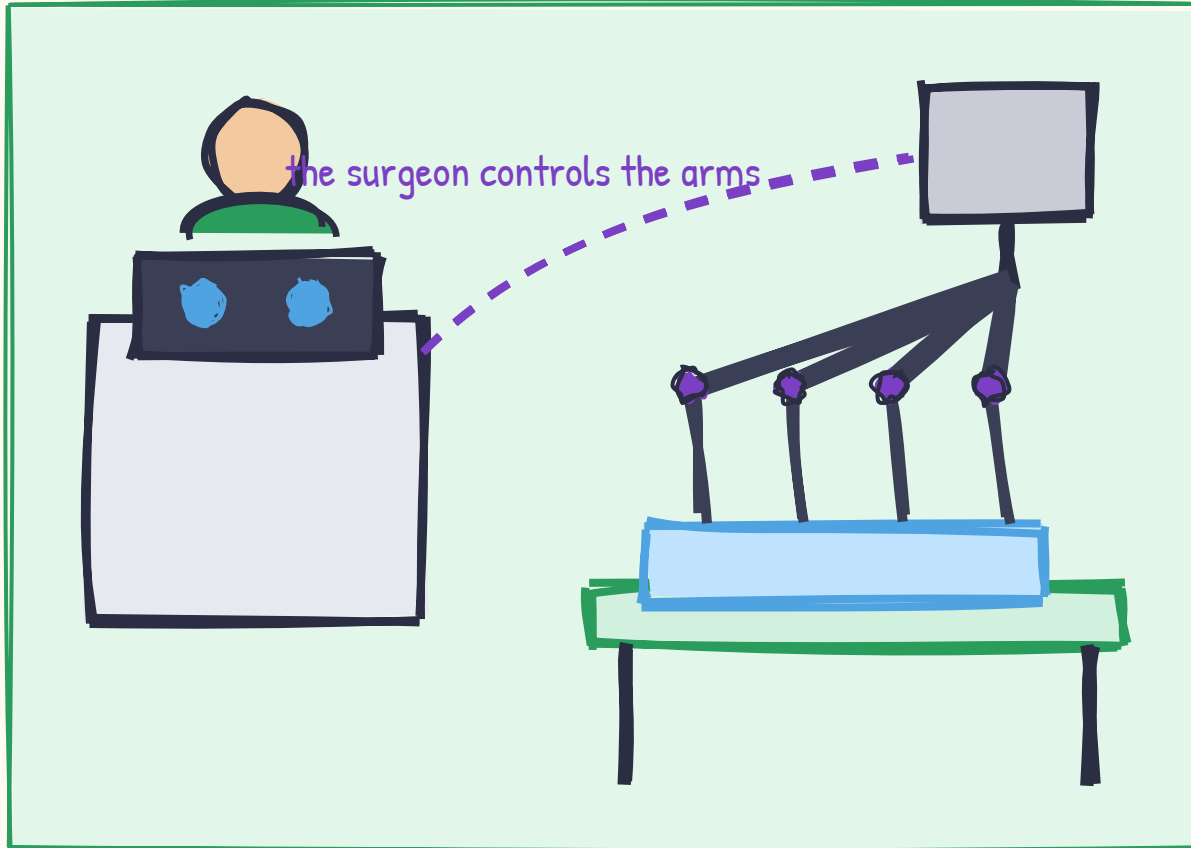
## Fascinating fact

Perseverance weighs about 1,000 kg and is powered by a nuclear battery, not solar panels.



## Surgical robot · Intuitive, USA

# da Vinci



### What it does

A robot that **helps doctors do operations** through tiny cuts. The surgeon sits at a console and moves controls; the robot's arms copy the movements with tiny tools.

**SENSE:** a 3D camera inside the body gives the surgeon a magnified view

**THINK:** turns big hand movements into tiny precise ones and filters out shaking

**ACT:** 4 robotic arms hold the camera and the instruments

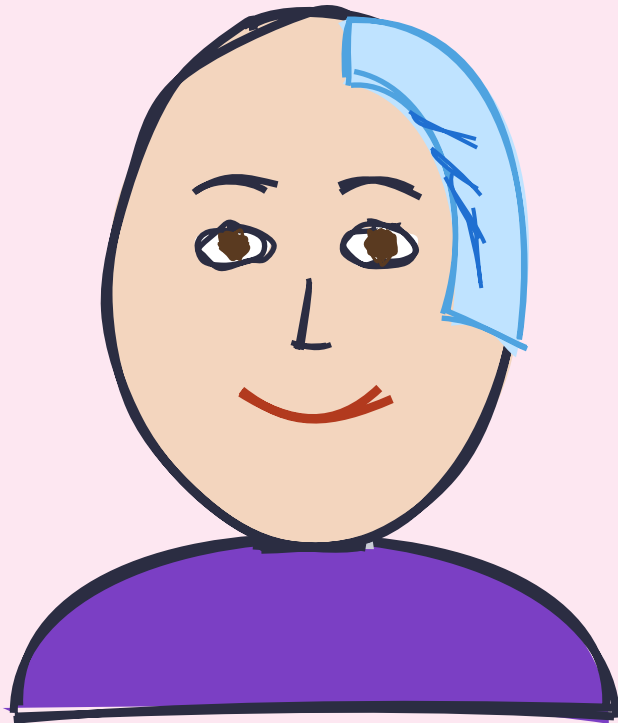
### Fascinating fact

Robots like this have helped with millions of operations around the world. The surgeon is always in control.



Social humanoid · Hanson Robotics, Hong Kong · 2016

# Sophia



## What it does

A robot with a **human-like face** that can smile, frown and hold simple conversations. She shows how robots might talk with people in the future.

**SENSE:** cameras in her eyes see faces; microphones hear speech

**THINK:** AI software chooses what to say and which expression to show

**ACT:** small motors under her skin move her face

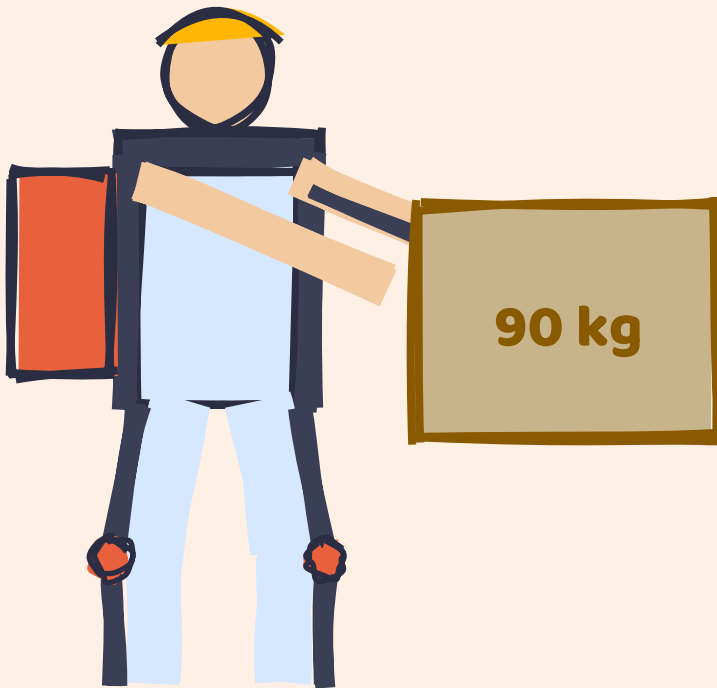
## Fascinating fact

In 2017 Sophia became the first robot to be given citizenship of a country (Saudi Arabia).



A wearable robot · real-life super suit

# Exoskeleton



## What it does

A robot you **wear**, like a metal skeleton outside your body. Factory workers use them to lift heavy loads, and medical ones help people who cannot walk to stand and walk again.

**SENSE:** sensors feel how the wearer moves and how heavy the load is

**THINK:** decides how much extra force to add, many times a second

**ACT:** motors at the hips, knees and shoulders add strength

## Fascinating fact

Some industrial exoskeletons let a worker lift around 90 kg again and again, with the robot taking most of the weight.



Moving robots · e.g. Amazon fulfilment centres

# Warehouse robots



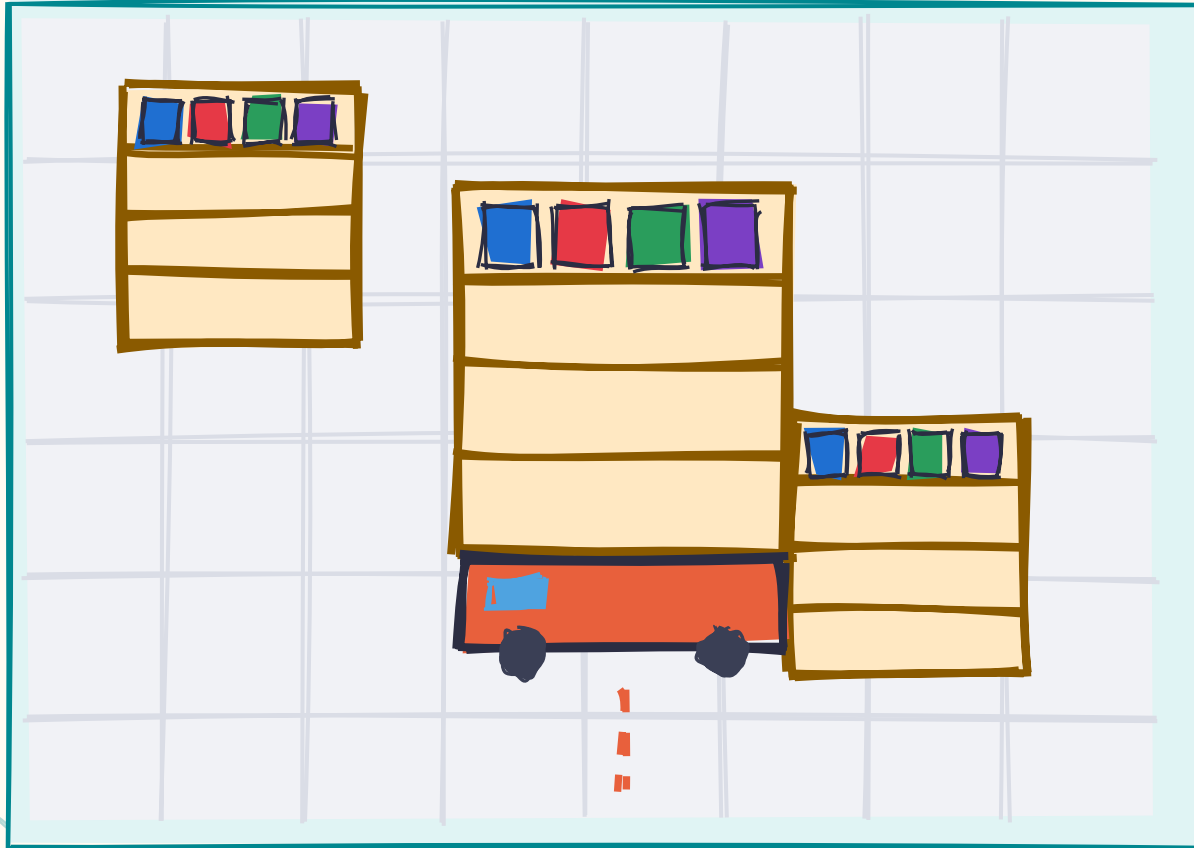
## What it does

Hundreds of small robots **drive under shelves**, **lift them and carry them** to workers who pack orders. They zoom around without crashing into each other.

**SENSE:** cameras read codes stuck on the floor to know exactly where they are  
**THINK:** a central computer plans routes for every robot at once  
**ACT:** wheels drive them; a lifting plate raises the shelf

## Fascinating fact

Amazon uses around one million robots in its warehouses, working side by side with people.



Inside every robot

# The 6 parts of a robot



**Sensors** = the eyes, ears and skin  
Collect information



**Controller** = the brain  
Runs the program, makes decisions



**Actuators** = the muscles  
Move, light up, make sound



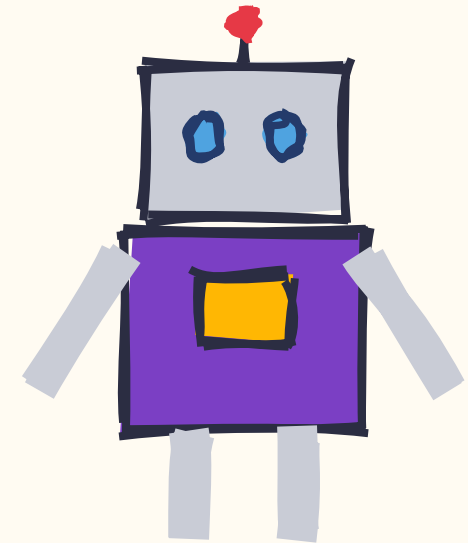
**Power** = the food and energy  
Batteries or a power supply



**Body** = the skeleton  
Holds everything together



**Program** = the thoughts  
Instructions written by us



Human vs robot

We have the same 6 parts! Our nerves are the wires.



## Sensing



# Sensors: how robots notice the world

**Ultrasonic**

Distance, using sound echoes

**Light (LDR)**

How bright or dark it is

**Camera**

Pictures, faces, colours

**Touch / button**

Being pressed or bumped

**Temperature**

How hot or cold it is

**Motion (PIR)**

A person moving nearby



Today we use the **ultrasonic sensor**. It shouts a sound too high for us to hear and listens for the **echo**, just like a bat.

**Your sensors**

Eyes, ears, nose, tongue, skin. Which one is like a camera?





Acting

# Actuators: how robots move and react

**DC motor**

Spins wheels and fans

**Servo motor**

Turns to an exact angle

**LED**

Lights up to show a signal

**Buzzer**

Beeps an alarm or a tune

**Screen**

Shows words and numbers

**Pump**

Moves water



Today our robot uses two actuators: a **red LED** to warn your eyes and a **buzzer** to warn your ears.

**Think**

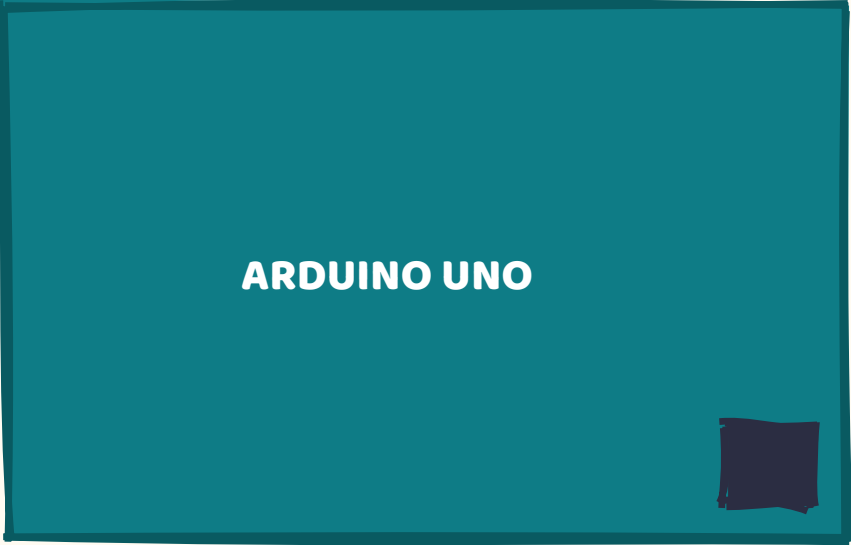
Which actuators does a robot vacuum need?





Thinking

# The brain: a microcontroller



ARDUINO UNO

## Bigger brain

Later we use the ESP32: a brain with Wi-Fi and Bluetooth built in!

### What is it?

A **microcontroller** is a tiny computer on a board. The **Arduino Uno** reads sensors, runs our program and controls actuators. It is cheap, safe (5 volts) and used by millions of learners.



### How do we program it?

1. Write the program on a laptop.
2. **Upload** it through a USB cable.
3. The Arduino runs it **by itself**, even without the laptop.



## Robots + AI

# When robots learn

### A normal robot

Follows **rules we write**.

*IF something is closer than 20 cm, THEN beep.*

### A robot with AI

Learns from **examples**.

Show it 1,000 pictures of cats and it learns to *recognise* a cat by itself.



Face unlock on phones



Cars that read road signs



Farm robots that pick ripe fruit



Voice assistants that understand you



Stand up and play!

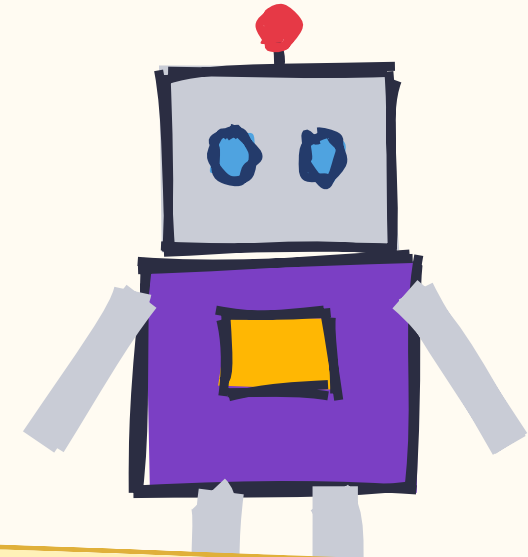
# Program a human robot

## How to play (pairs)

1. One partner is the **robot**, one is the **programmer**.
2. The robot must reach a target (a chair, a door) and pick up an object.
3. The robot follows instructions **exactly**, nothing more: "step forward", "turn right", "stop".
4. If the program is wrong, the robot does the wrong thing!



Was your program right the first time? Fixing mistakes in a program is called **debugging**.



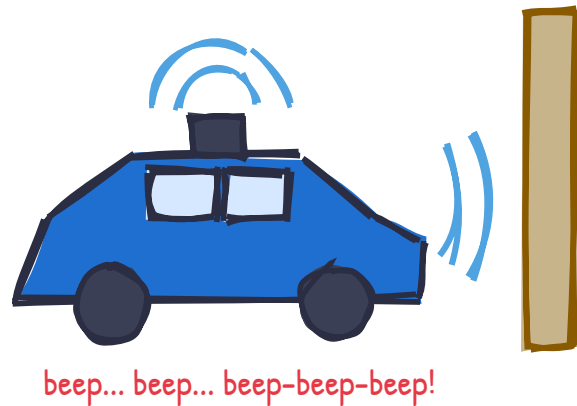
### Robot rule

Robots do what we TELL them, not what we MEAN!



Your first build

# Make a Distance Alarm robot



## Like a car parking sensor

When a car reverses towards a wall, it **beeps** faster and faster. Today you build the same thing: when something comes **closer than 20 cm**, a red light turns on and a buzzer beeps.

## Sense, think, act

**Sense:** ultrasonic sensor measures distance

**Think:** Arduino checks: closer than 20 cm?

**Act:** LED on + buzzer beeps



Build it

# Our builder's kit (per pair)



**Arduino Uno + USB**

The brain



**HC-SR04 sensor**

The eyes (ultrasonic)



**Red LED + 220  $\Omega$**

Light alarm



**Active buzzer**

Sound alarm



**Breadboard + wires**

Roads for electricity



**Laptop**

Upload the program



## Two ways to build

**Ready-built:** the teacher has wired it. Plug in, upload, test and change the program.

**Build it yourself:** follow the wiring guide on the next slide, then ask the teacher to check before you plug in.

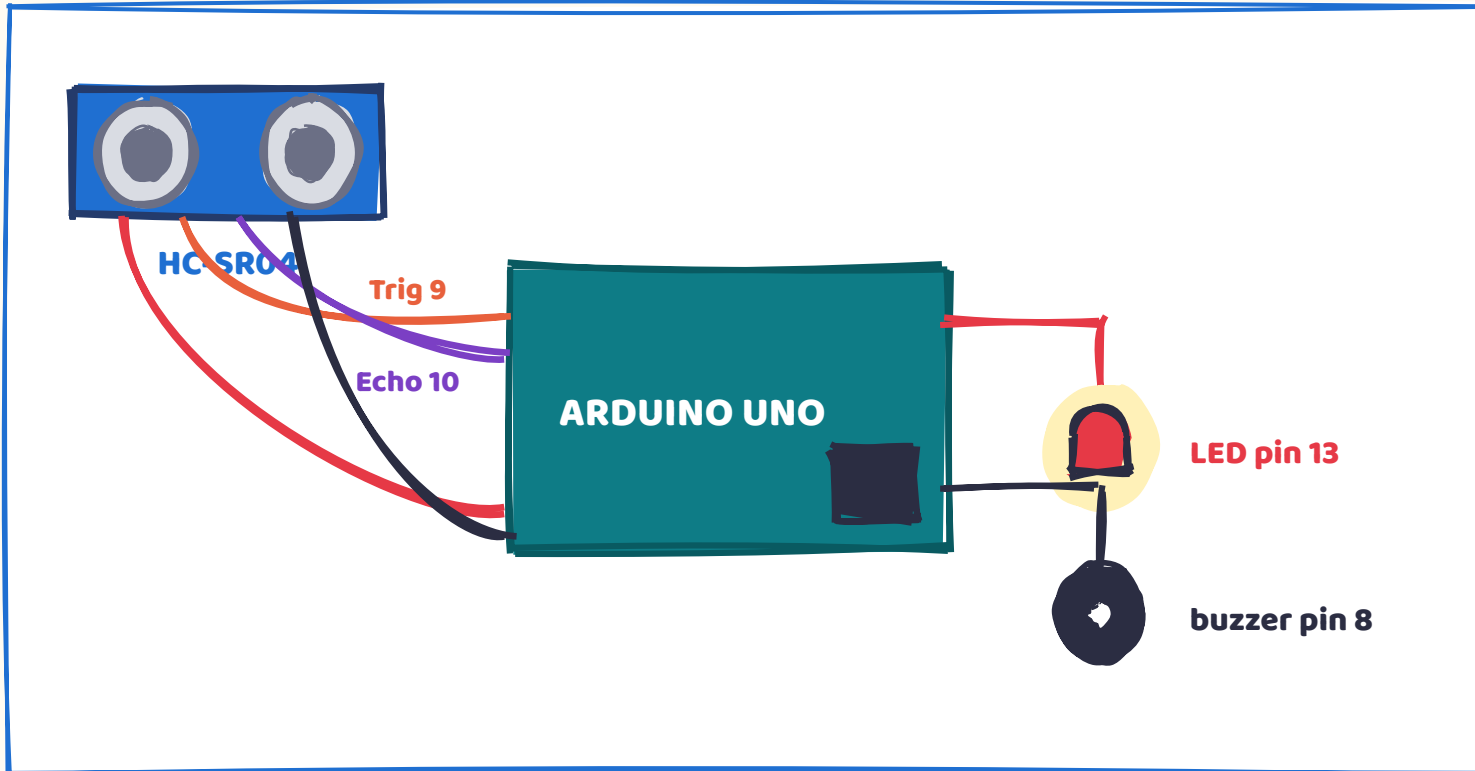
## Safety first

Always unplug the USB cable before you change any wires.



Build it

# Wiring guide



| Part                           | Arduino |
|--------------------------------|---------|
| Sensor VCC                     | 5V      |
| Sensor Trig                    | pin 9   |
| Sensor Echo                    | pin 10  |
| Sensor GND                     | GND     |
| LED long leg (+ 220 $\Omega$ ) | pin 13  |
| Buzzer +                       | pin 8   |
| LED short leg, buzzer -        | GND     |

### Check twice

Red wire to 5V, black wire to GND. Swapping them can damage the sensor!



## The program

# Ready-made code: just upload!



```
// Distance Alarm: like a car parking sensor
const int TRIG = 9, ECHO = 10, LED = 13, BUZZ = 8;
int ALARM_CM = 20;           // <-- change me!

void setup() {
  pinMode(TRIG, OUTPUT);  pinMode(ECHO, INPUT);
  pinMode(LED, OUTPUT);   pinMode(BUZZ, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  digitalWrite(TRIG, HIGH); delayMicroseconds(10);
  digitalWrite(TRIG, LOW);           // SENSE: send a sound pulse
  long cm = pulseIn(ECHO, HIGH, 30000) / 58;
  Serial.println(cm);                // see the distance

  if (cm > 0 && cm < ALARM_CM) {      // THINK: too close?
    digitalWrite(LED, HIGH);         // ACT: light on
    digitalWrite(BUZZ, HIGH);        // and beep
  } else {
    digitalWrite(LED, LOW);
    digitalWrite(BUZZ, LOW);
  }
  delay(100);
}
```

### Find the 3 parts

**SENSE:** send a sound and time the echo.

**THINK:** if the distance is less than ALARM\_CM...

**ACT:** turn the LED and buzzer on.



### Your only job

Change ALARM\_CM from 20 to another number. Upload again. What changes?

Teacher: the file is shared on the lab laptops. Open the Serial Monitor (9600) to see live distances.



Test like an engineer

# Does your robot sense correctly?



| Test                      | Expect              | Reading  | Worked?  |
|---------------------------|---------------------|----------|----------|
| Hand 50 cm away           | LED off, quiet      | _____ cm | yes / no |
| Hand 10 cm away           | LED on, beeping     | _____ cm | yes / no |
| A book instead of a hand  | Does it still work? | _____ cm | yes / no |
| A soft cushion or sweater | Does it still work? | _____ cm | yes / no |

## Not working?

Always 0? Trig and Echo are swapped.  
LED never lights? Long leg to pin 13?  
No beep? Buzzer + to pin 8?  
Nothing at all? Is the USB plugged in?



Why might a soft cushion fool the sensor? (Hint: what happens to sound when it hits something soft?)



Level up!

# Choose your challenge



## Level 1

Change the alarm distance to 10 cm, then to 50 cm.



## Level 2

Add a green LED that is ON when everything is safe.



## Level 3

Make it beep faster as things get closer, just like a real car.

```
// Level 3 hint
digitalWrite(BUZZ, HIGH);
delay(50);
digitalWrite(BUZZ, LOW);
delay(cm * 10); // closer = shorter gap
```



### Real-world idea

Where else could a distance alarm help? A blind person's cane? A garage door?





Coming up in the club

# What will we build next?



## Robot cars

that avoid walls and follow lines



## Robot arms

that copy your hand



## Smart devices

you control from your phone



## Smart farms

that water plants by themselves



## Safety systems

for fire, gas and security



## AI projects

that see and recognise things



Robotics jobs: **robotics engineer**, **AI engineer**, **drone pilot**, **electronics designer**, **programmer**... and jobs that do not even exist yet!





Show what you know

# What did we learn?



A robot senses, thinks and acts by itself



Robots have sensors, a controller, actuators, power, a body and a program



Robots work in homes, factories, hospitals and even on Mars



An ultrasonic sensor measures distance with echoes



A program tells the robot exactly what to do



**Robot Builder**

badge unlocked!



**At home**

Count the robots and sensors you can find at home.  
Bring your list next class!



**Exit ticket:** name one SENSOR and one ACTUATOR in your Distance Alarm.

